VI SEMESTER (PROFESSIONAL ELECTIVE-II)	L	T	P	C
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20AM6E02 - ETHICS OF ARTIFICIAL INTELLIGENCE				

COURSE OBJECTIVES:

1. To understand the need for ensuring ethics in AI.
2. To understand ethical issues with the development of AI agents.
3. To apply the ethical considerations in different AI applications.
4. To evaluate the relation of ethics with nature.
5. To overcome the risk for Human rights and other fundamental values.

COURSE OUTCOMES:

CO1: Understand the ethical issues in the development of AI agents

CO2: Learn the ethical considerations of AI with perspectives on ethical values

CO3: Apply the ethical policies in AI based applications and Robot development

CO4: To implement the AI concepts to societal problems by adapting the legal concepts by securing fundamental rights.

CO5: This study will help to overcome the evil genesis in the concepts of AI.

UNIT-I: INTRODUCTION TO ETHICS OF AI:

Role of Artificial Intelligence in Human Life, Understanding Ethics, Why Ethics in AI? Ethical Considerations of AI, Current Initiatives in AI and Ethics, Ethical Issues with our relationship with artificial Entities

UNIT -II: FRAMEWORK AND MODELS:

AI Governance by Human-right centered design, Normative models, Role of professional norms, Teaching Machines to be Moral

UNIT -III: CONCEPTS AND ISSUES:

Accountability in Computer Systems, Transparency, Responsibility and AI.Race and Gender, AI as a moral right-holder.

UNIT -IV: PERSPECTIVES AND APPROACHES:

Perspectives on Ethics of AI, Integrating ethical values and economic value, Automating origination, AI a Binary approach, Machine learning values, Artificial Moral Agents.

UNIT -V: CASES AND APPLICATION:

Ethics of Artificial Intelligence in Transport, Ethical AI in Military, Biomedical research, Patient Care, Public Health, Robot Teaching, Pedagogy, Policy, Smart City Ethics.

Reference Books:

1. Paula Boddington, “Towards a Code of Ethics for Artificial Intelligence”, Springer, 2017
2. Markus D. Dubber, Frank Pasquale, Sunit Das, “The Oxford Handbook of Ethics of AI”, Oxford University Press Edited book, 2020
3. S. Matthew Liao, “Ethics of Artificial Intelligence”, Oxford University Press EditedBook, 2020
4. N. Bostrom and E. Yudkowsky.“The ethics of artificial intelligence”. In W. M. Ramsey and K. Frankish, editors, The Cambridge Handbook of Artificial Intelligence, pages 316–334. Cambridge University Press, Cambridge, 2014.
6. Wallach, W., & Allen, C, “Moral machines: ceaching robots right from wrong”, Oxford University Press, 2008.